Select one of the following Multi-Agent Projects.

- In teams of 2 - no exceptions!
- Demonstrate understanding of multi-agent architectures, LLM integration, prompt engineering, and system design.
- Showcasing different aspects of multi-agent systems: collaboration, specialization, strategic reasoning, and coordination.

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1. Multi-Agent Debate System

Concept: Build a system where two AI agents debate a topic while a moderator agent manages the discussion and a judge agent evaluates arguments.

Agents (4 total):

- **Moderator Agent**: Introduces topic, enforces rules, manages turn-taking
- **Debater Agent A**: Argues for the proposition
- **Debater Agent B**: Argues against the proposition
- **Judge Agent**: Evaluates arguments, identifies fallacies, declares winner

Implementation (~40 hours):

- Basic agent framework and communication (8 hours)
- Agent prompt engineering and personality design (10 hours)
- Debate flow logic and turn management (8 hours)
- Scoring/evaluation system (6 hours)
- UI and testing (8 hours)

Deliverables: Working demo with 3-5 debate topics, evaluation rubric, documentation of agent prompts and interaction patterns

Why it works: Clear agent roles, manageable scope, demonstrates key concepts (agent communication, role specialization, evaluation), impressive demo potential

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2. Collaborative Story Writing System

Concept: Multiple specialized agents work together to write a coherent short story.

Agents (3-4 total):

- **Plot Agent**: Creates story outline and maintains narrative arc
- **Character Agent**: Develops consistent character personalities and dialogue
- **Scene Agent**: Writes descriptive scenes and action sequences
- **Editor Agent**: Ensures consistency, pacing, and quality

Implementation (~40 hours):

- Agent architecture and state management (8 hours)
- Individual agent prompt design and testing (12 hours)
- Story coherence and memory system (10 hours)
- Iteration/refinement logic (6 hours)
- Interface and output formatting (4 hours)

Deliverables: System that generates 1000-1500 word stories in chosen genres, demonstrates revision capability, analysis of agent contributions

Why it works: Creative and engaging, clear evaluation criteria (story quality, coherence, character consistency), natural agent specialization, good for demonstrating memory/context management

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3. Multi-Agent Code Review System

Concept: Agents collaborate to review code submissions, identifying different types of issues and suggesting improvements.

Agents (4 total):

- **Security Agent**: Scans for security vulnerabilities and unsafe practices
- **Performance Agent**: Identifies optimization opportunities and inefficiencies
- **Style Agent**: Checks code style, readability, and best practices
- **Integration Agent**: Synthesizes feedback and prioritizes suggestions

Implementation (~40 hours):

- Agent framework and code parsing (8 hours)
- Individual agent logic and rule sets (12 hours)
- Feedback aggregation and prioritization (8 hours)
- Report generation system (6 hours)
- Testing with sample codebases (6 hours)

Deliverables: System that reviews code files in 2-3 languages, consolidated review report, comparison with static analysis tools

Why it works: Practical application, clear measurable outcomes, demonstrates agent specialization, easy to evaluate objectively

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4. Multi-Agent Customer Support Simulation

Concept: Simulate a customer support system where agents handle different aspects of customer inquiries.

Agents (4 total):

- **Triage Agent**: Categorizes inquiries and determines urgency
- **Knowledge Base Agent**: Searches documentation and FAQs for solutions
- **Specialist Agent**: Handles complex technical issues
- **Follow-up Agent**: Ensures issue resolution and customer satisfaction

Implementation (~40 hours):

- Agent routing and communication logic (8 hours)
- Knowledge base integration (simple vector store) (10 hours)
- Conversation flow management (10 hours)
- Response quality and escalation logic (6 hours)
- UI and conversation logging (6 hours)

Deliverables: System handling 5-10 support scenarios, conversation transcripts showing agent handoffs, metrics on resolution rates

Why it works: Real-world applicability, demonstrates agent coordination and handoffs, good for showing decision-making logic

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5. Multi-Agent News Article Fact-Checker

Concept: Agents collaborate to verify claims in news articles and assess their credibility.

Agents (4 total):

- **Claim Extraction Agent**: Identifies factual claims in articles
- **Research Agent**: Searches for corroborating or contradicting sources
- **Evidence Evaluation Agent**: Assesses source credibility and evidence quality
- **Verdict Agent**: Synthesizes findings into fact-check ratings

Implementation (~40 hours):

- Article parsing and claim extraction (8 hours)
- Web search integration and source retrieval (10 hours)
- Evidence assessment logic (10 hours)
- Verdict synthesis and explanation (6 hours)
- Report generation and UI (6 hours)

Deliverables: System that fact-checks 3-5 articles, detailed reports with sources, confidence scores for each claim

Why it works: Highly relevant topic, clear evaluation metrics, demonstrates research coordination, practical societal value

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6. Multi-Agent Meeting Assistant

Concept: Agents process meeting transcripts to extract different types of information and generate comprehensive summaries.

Agents (4 total):

- **Action Items Agent**: Identifies tasks, owners, and deadlines
- **Decision Tracker Agent**: Extracts key decisions and rationale
- **Topic Summarizer Agent**: Creates summaries of discussion topics

- **Attendee Insights Agent**: Tracks participation and sentiment

Implementation (~40 hours):

- Transcript processing and speaker identification (6 hours)
- Individual agent extraction logic (14 hours)
- Cross-agent information reconciliation (8 hours)
- Structured output generation (6 hours)
- Testing with sample meetings (6 hours)

Deliverables: System processing 5-10 meeting transcripts, structured outputs (action items, decisions, summaries), comparison with single-agent approach

Why it works: Clear business value, easy to evaluate outputs, demonstrates parallel processing, relatable use case

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7. Multi-Agent Academic Paper Reviewer System

Concept: Agents simulate peer review process by evaluating research papers from different perspectives.

Agents (4 total):

- **Methodology Agent**: Evaluates research methods and experimental design
- **Novelty Agent**: Assesses originality and contribution to field
- **Clarity Agent**: Reviews writing quality, structure, and presentation
- **Meta-Reviewer Agent**: Synthesizes reviews and makes recommendations

Implementation (~40 hours):

- PDF/paper parsing and section identification (6 hours)
- Individual reviewer agent logic (14 hours)
- Scoring rubric implementation (8 hours)
- Meta-review synthesis (6 hours)
- Review report generation (6 hours)

Deliverables: System reviewing 3-5 papers, detailed reviews with scores, comparison with actual peer reviews (if available)

Why it works: Academic relevance, structured evaluation criteria, shows consensus-building, validates against real reviews

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8. Multi-Agent Job Application Optimizer

Concept: Agents collaborate to help candidates optimize their job applications for specific positions.

Agents (4 total):

- **Job Analysis Agent**: Extracts key requirements and qualifications from job posting
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- **Resume Optimizer Agent**: Tailors resume to highlight relevant experience
- **Cover Letter Agent**: Writes personalized cover letter
- **Interview Prep Agent**: Generates likely interview questions and talking points

Implementation (~40 hours):

- Job posting and resume parsing (8 hours)
- Requirement matching algorithm (8 hours)
- Resume tailoring logic (8 hours)
- Cover letter and interview prep generation (10 hours)
- Output formatting and comparison view (6 hours)

Deliverables: System processing 5+ job applications, before/after comparisons, match scores, interview preparation materials

Why it works: Personal relevance to students, clear before/after evaluation, demonstrates information synthesis, practical career value

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9. Multi-Agent Legal Document Analyzer

Concept: Agents collaborate to analyze contracts or legal documents, identifying risks, obligations, and key terms.

Agents (4 total):

- **Clause Extraction Agent**: Identifies and categorizes contract clauses
- **Risk Assessment Agent**: Flags problematic terms and legal risks
- **Obligation Tracker Agent**: Extracts parties' responsibilities and deadlines
- **Plain Language Agent**: Translates legal jargon into understandable summaries

Implementation (~40 hours):

- Document parsing and section identification (8 hours)
- Clause classification and extraction (10 hours)
- Risk detection rules and logic (8 hours)
- Plain language translation (8 hours)
- Summary report generation (6 hours)

Deliverables: System analyzing 5-10 sample contracts (NDAs, employment agreements), risk reports, obligation timelines, readability comparison

Why it works: High practical value, clear evaluation criteria, demonstrates specialized analysis, addresses real pain point

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10. Multi-Agent Recipe and Meal Planner

Concept: Agents work together to create personalized meal plans considering nutrition, preferences, budget, and cooking time.

Agents (4 total):

- **Nutritionist Agent**: Ensures balanced macros/micros and dietary requirements
- **Budget Agent**: Optimizes for cost and ingredient overlap
- **Recipe Curator Agent**: Selects or adapts recipes based on constraints
- **Scheduler Agent**: Organizes meal prep timeline and shopping lists

Implementation (~40 hours):

- Recipe database setup (simple JSON or API) (6 hours)
- Individual agent constraint logic (14 hours)
- Multi-objective optimization (8 hours)
- Meal plan generation and adjustment (6 hours)
- Shopping list and schedule output (6 hours)

Deliverables: Weekly meal plans for 3-5 user profiles, nutritional analysis, cost breakdowns, shopping lists

Why it works: Relatable problem, measurable outcomes (nutrition, cost), demonstrates constraint satisfaction, creative agent coordination

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11. Multi-Agent Research Paper Summarizer

Concept: Agents extract and synthesize different aspects of research papers to create comprehensive multi-level summaries.

Agents (4 total):

- **Methods Agent**: Extracts and explains research methodology
- **Results Agent**: Identifies key findings and data
- **Context Agent**: Explains background, motivation, and related work
- **Impact Agent**: Assesses implications and future research directions

Implementation (~40 hours):

- Paper parsing (PDF/text extraction) (6 hours)
- Section identification and routing (6 hours)
- Individual agent summarization logic (12 hours)
- Multi-level summary synthesis (executive, detailed, technical) (10 hours)
- Citation and reference handling (6 hours)

Deliverables: System summarizing 5-8 papers, summaries at multiple depths, comparison with abstract, key findings extraction

Why it works: Academic relevance, validates against existing abstracts, demonstrates information hierarchy, useful for literature reviews

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12. Multi-Agent Personal Finance Advisor

Concept: Agents analyze financial data and provide comprehensive advice from different financial planning perspectives.

Agents (4 total):

- **Spending Analyzer Agent**: Categorizes expenses and identifies patterns
- **Savings Optimizer Agent**: Recommends budget adjustments and savings strategies
- **Investment Advisor Agent**: Suggests investment allocations based on goals/risk
- **Goal Tracker Agent**: Monitors progress toward financial objectives

Implementation (~40 hours):

- Transaction data parsing (CSV/synthetic data) (6 hours)
- Spending categorization and analysis (8 hours)
- Recommendation engine logic (10 hours)
- Goal setting and tracking (8 hours)
- Dashboard and visualization (8 hours)

Deliverables: System analyzing sample financial data, personalized recommendations, savings projections, goal timelines, what-if scenarios

Why it works: Universal relevance, quantifiable results, demonstrates reasoning with numerical data, actionable outputs

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